

Team Qwerty

3D CT-Based Pulmonary Nodule Detection using Deep Learning

An AI Assisted Framework for Early Lung Cancer Screening

97.67%

Avg Accuracy

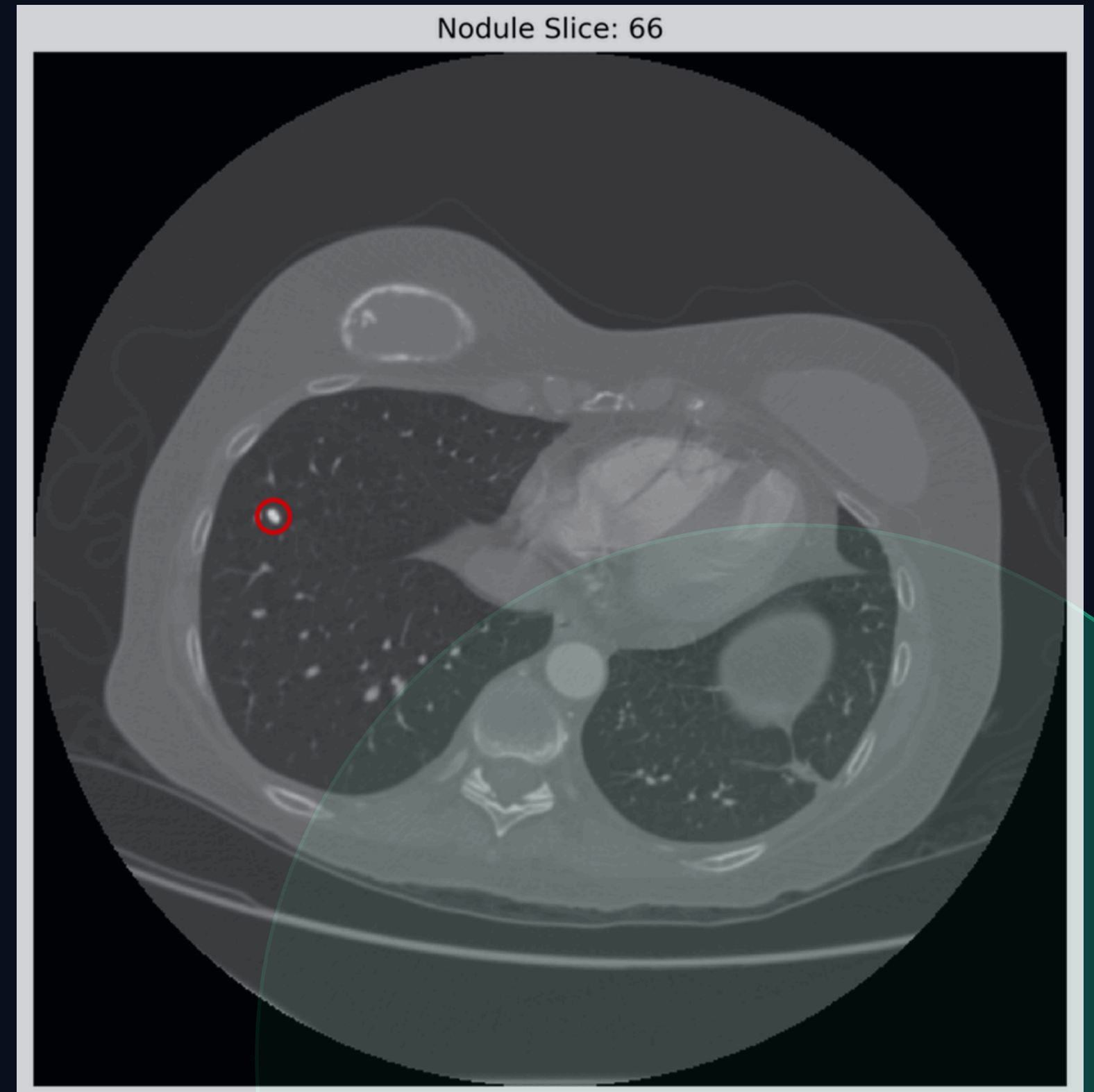
0.9954

ROC-AUC

LUNA16

Dataset

*Angelica Das(230103014) | Kasukurthi Ananya(230106034)
| Yash Mohan(230122065)*



Why Early Lung Nodule Detection Matters

Problem Statement

01 Leading Cause of Cancer Mortality

Lung cancer ranks among the top causes of cancer-related deaths worldwide.

02 Silent Early Stage Nodules

Early pulmonary nodules are often clinically silent yet indicate potential malignancy.

03 Manual CT Examination Limitations

Volumetric CT review is time consuming, error-prone, and susceptible to radiologist fatigue.

04 Detection Complexity

Small nodules in dense 3D scans are difficult to reliably identify by the human eye alone.

Project Objective

What LungNoduleNet sets out to achieve

Core Objective

Develop a robust 3D Convolutional Neural Network (3D-CNN) for automated pulmonary nodule detection from volumetric chest CT scans, enabling faster and more reliable early lung cancer screening.

Improve Sensitivity

Detect more true nodules, reducing missed cases in high-risk patients.

Reduce False Positives

Minimize unnecessary follow-up procedures through precise classification.

Radiology Assistance

Support radiologists with AI driven second opinion screening tools.

Dataset: LUNA16

LUNG Nodule Analysis 2016 — The Gold Standard Benchmark

888

Low-Dose CT Scans

4,032

Total Samples

1,086

Positive Nodules

2,946

Negative Samples

Radiologist-Annotated

Expert verified nodule coordinates and diagnostic labels across all scans.

Full Dataset Used

LUNA16 Subsets 0–9 all utilized for robust volumetric learning.

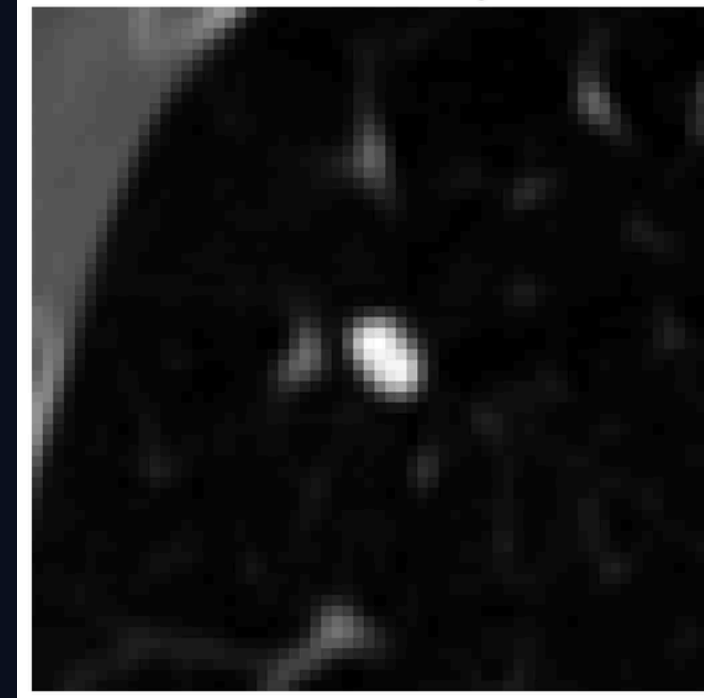
Normalized Cubes

3D patches of shape (1, 64, 64, 64) extracted per candidate coordinate.

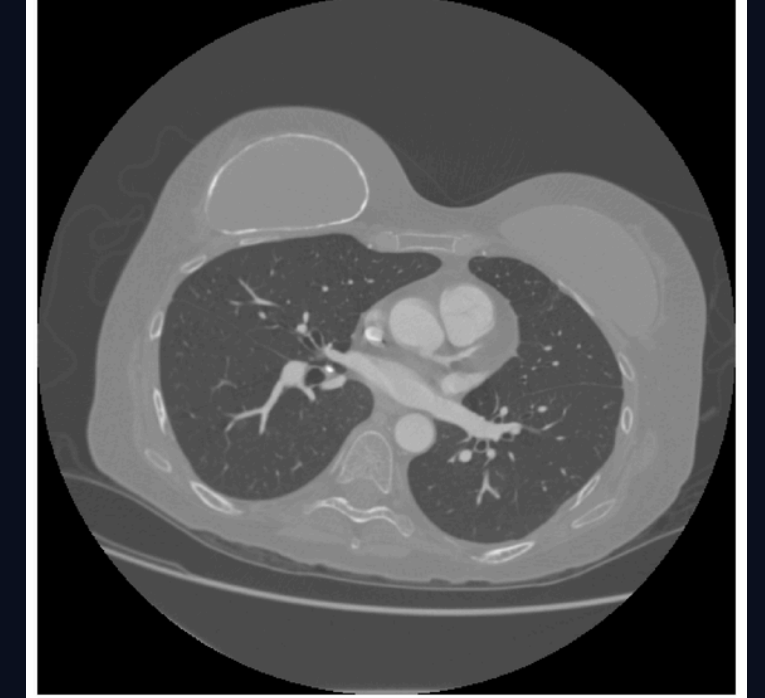
Dataset Shape

Raw: (4032, 64, 64, 64) · Labels: (4032,) · Normalized: (4032, 1, 64, 64, 64)

Extracted Nodule Region



Middle Slice: 116



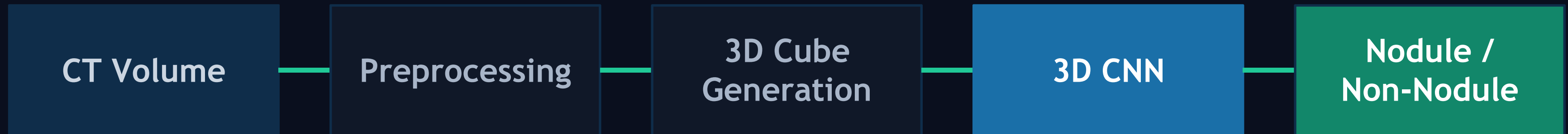
Raw dataset shape: (4032, 64, 64, 64)

Labels shape: (4032,)

Normalized shape: (4032, 1, 64, 64, 64)

LungNoduleNet Pipeline & Architecture

End-to-end 3D CNN workflow from CT volume to nodule prediction



3D-CNN Architecture

Conv3D → BatchNorm → ReLU → MaxPool3D

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Conv3D → BatchNorm → ReLU → MaxPool3D

Flatten → Fully Connected Layer

Softmax → Nodule / Non-Nodule

Why 3D CNN?



Captures spatial depth from volumetric CT ; missed entirely by conventional 2D CNNs.

Training Configuration

Hyperparameters, framework & hardware setup

Validation

10-Fold Cross Validation

Epochs

18

Optimizer

Adam

Learning Rate

1e-4

Loss Function

Weighted Cross Entropy

Framework

PyTorch

Hardware

Tesla T4 × 2

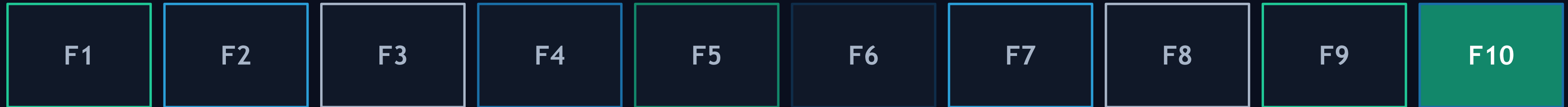
LR Scheduling

Stable Convergence

Learning rate scheduling applied for stable convergence across all 18 training epochs.

10-Fold Cross Validation Strategy

Robust evaluation across all data splits



Better Generalization

Trains & tests on every subset, yielding a model that generalizes across unseen CT scans.

Reduced Bias

Each fold serves once as the test set, eliminating selection bias in the evaluation.

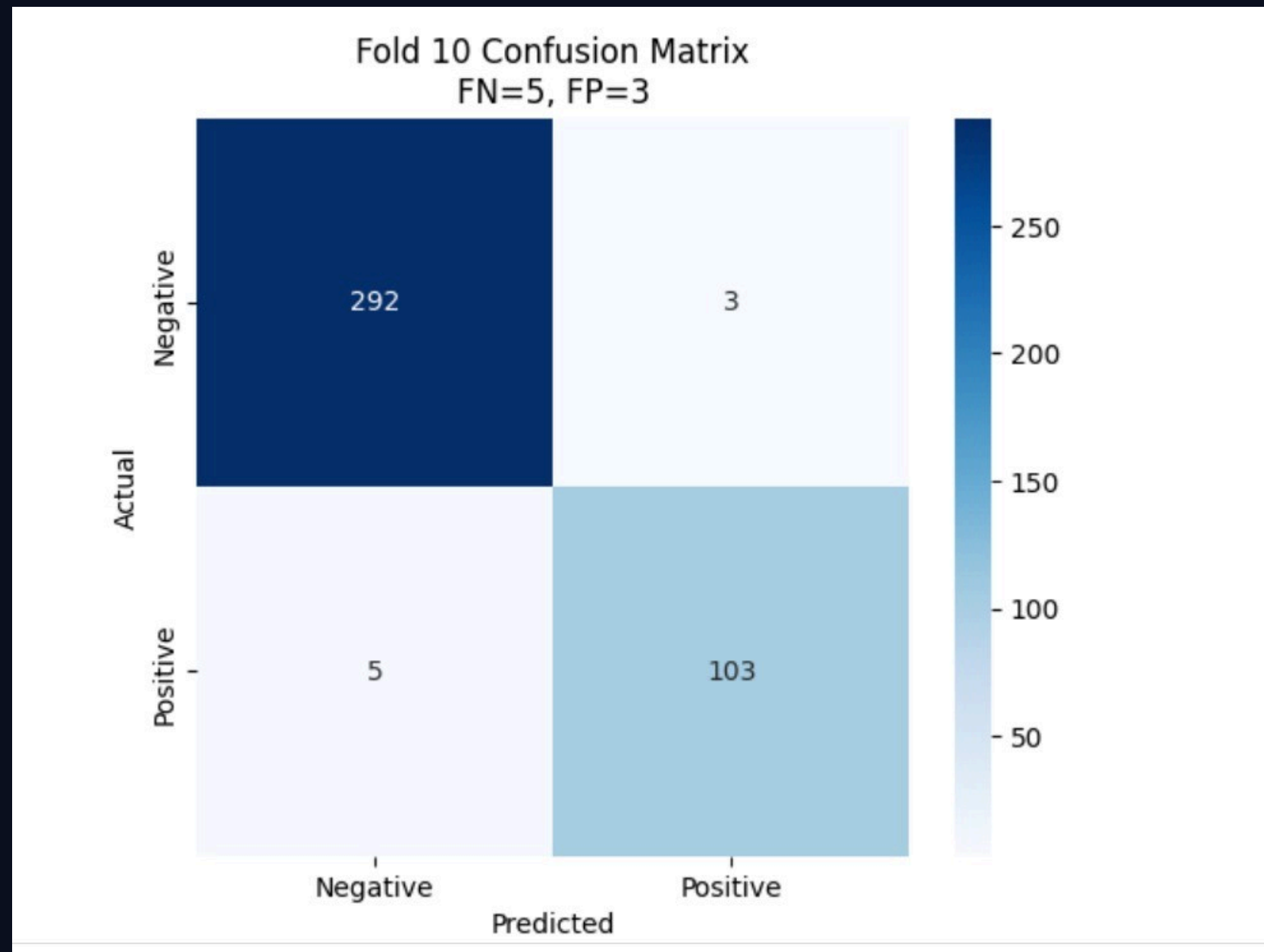
Reliable Evaluation

Averaged accuracy across 10 folds gives a statistically robust performance estimate.

Average 10-Fold Accuracy: 97.66896788934477

Fold 10 Confusion Matrix

Classification outcomes : nodule vs. non-nodule



292

True Negatives

Correctly identified non-nodule cases

103

True Positives

Correctly detected nodule cases

5

False Negatives

Missed nodules (critical to minimize)

3

False Positives

Incorrect nodule flags

Fold 10 Classification Report

Precision · Recall · F1-Score · Accuracy

97.17%

Precision

95.37%

Recall / Sensitivity

96.26%

F1-Score

98.01%

Accuracy

Fold 10 Classification Report

	precision	recall	f1-score	support
Negative	0.9832	0.9898	0.9865	295
Positive	0.9717	0.9537	0.9626	108
accuracy			0.9801	403
macro avg	0.9774	0.9718	0.9746	403
weighted avg	0.9801	0.9801	0.9801	403

Clinical Insight

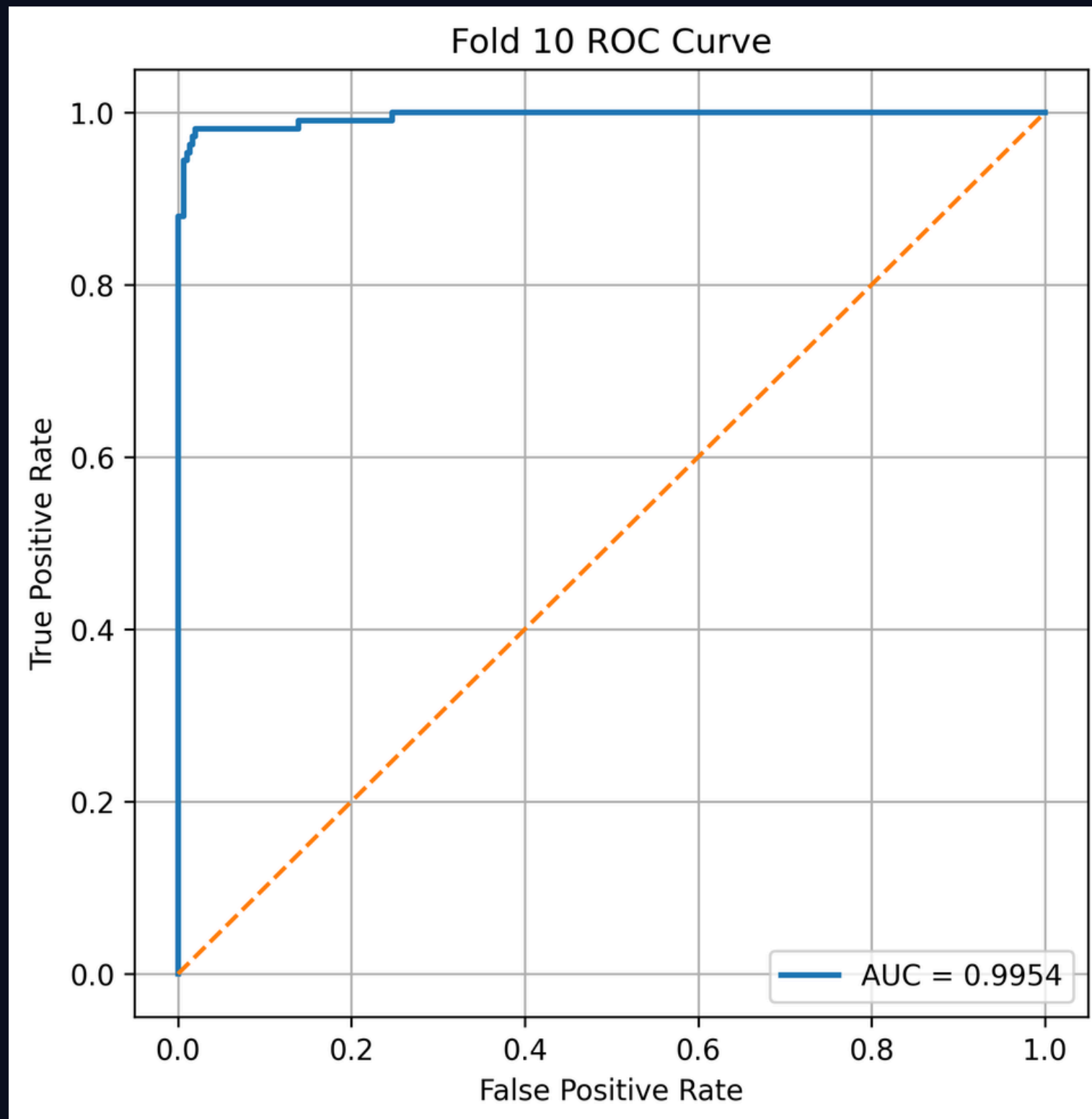
High recall (95.37%) ensures fewer missed nodules ; the most clinically critical metric for cancer screening.

High precision (97.17%) minimizes false alarms, reducing unnecessary patient distress.

The balance between both metrics results in a strong F1-Score of 96.26%.

ROC-AUC Performance

Near-perfect class separability between nodule and non-nodule



AUC

0.9954

What this means

An AUC of 0.9954 indicates near perfect discrimination between nodule and non nodule cases.

The model's ROC curve hugs the top left corner — a hallmark of clinical grade classifier performance.

Limitations & Future Scope

Where we are today and where LungNoduleNet goes next

Current Limitations

- Binary classification only , nodule vs. non-nodule
- No malignancy grading or benign/malignant prediction
- No clinical metadata fusion (patient age, history)
- Limited to LUNA16 data; no external clinical validation

Future Scope

- Full clinical deployment with hospital PACS integration
- Transformer based 3D architectures (e.g. ViT-3D)
- PET scan & clinical data fusion for richer context
- Real-time Computer Aided Detection (CAD) assistance

C O N C L U S I O N

LungNoduleNet

Demonstrates the effectiveness of 3D deep learning for pulmonary nodule detection using the complete LUNA16 dataset (Subsets 0–9)

97.67%

Average 10-Fold Accuracy

0.9954

ROC-AUC , Near Perfect

**FP=3
FN=5**

Low Error , Fold 10

LungNoduleNet establishes strong potential as an AI assisted tool for early lung cancer screening, with clinical deployment and multi-modal expansion as the natural next step.